



Achieving climate neutrality in district heating: The impact of system temperature levels on the supply mix of EU-27 in 2050

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ABSTRACT

District heating (DH) systems are a critical component of the EU's strategy to achieve climate neutrality in the heating sector by 2050. Lowering the system temperature levels in district heating systems is acknowledged as a key facilitator for integrating renewable energy sources, thereby advancing the decarbonisation of the DH sector. However, the impact of system temperature levels on a decarbonised pan-European level DH sector remains underexplored. This paper investigates the effect of system temperatures on the DH supply mix of EU-27 in 2050. Through an extensive scenario analysis, we show that reducing system temperature levels can reduce the levelized cost of heat generation by 20 %. We find that large-scale heat pumps and geothermal energy plants are key technologies, particularly effective in low-temperature DH systems. Conversely, wastewater heat pumps and biomass tend to be more favourable for high-temperature scenarios. Geothermal energy exhibits potential in low-temperature systems but faces barriers of high initial capital costs. The role of hydrogen is found to be marginal, and biomass's role is highly sensitive to its price. From a policy perspective, our analysis indicates that targeted investments and regulations are needed to accelerate the transition to low-temperature systems, enabling the incorporation of more sustainable technologies.

1. Introduction

The European Union's Climate Law and Green Deal have set an ambitious target to achieve carbon neutrality by 2050, placing the spotlight on the heating and cooling sector, which accounts for nearly half of the EU's total energy consumption [1]. DH could play a significant role in integrating low-carbon energy sources into the heating energy mix on a large scale [2]. However, two-thirds of the DH fuel supply mix in the EU-27 consists of fossil fuels, indicating an urgent need for systemic changes [3].

DH systems traditionally operate at high-temperature levels to ensure efficient heat transfer and distribution. However, recent advancements and environmental considerations have spotlighted the significance of reducing these temperature levels. Historically, DH systems have evolved through various generations: the 1st generation, which used steam as the heat carrier; the 2nd generation, which utilized high-temperature water; the 3rd generation, which introduced medium-

temperature water systems; and now, the 4th generation, focused on low-temperature systems that enhance energy efficiency and integrate renewable energy sources [4]. Emerging now is the 5GDH, which integrates district heating and cooling (DHC) systems, enabling even greater flexibility and efficiency by using ultra-low temperature networks and leveraging synergies between heating and cooling demands.

Ref. [4] gives an overview of the historical development of the past generations of DH and defines 4GDH, following the 3GDH. The DH generations differ in terms of system temperatures, heat carriers (steam, pressurised hot water, low-temperature water), type of pipes used (in situ/pre-insulated, steel/flexible), integration with smart energy systems, digitalisation, and many other aspects [4]. Even though the system temperature level is not the only characteristic of the different DH generations, it is one of the most fundamental differences among them.

The benefits of lowering system temperatures in DH are multifaceted. Reduced system temperatures increase efficiency on the heat production and network sides. Lower supply temperatures can increase the efficiency of not only heat pumps and solar thermal but also

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Abbreviations

3GDH	3rd generation of district heating
4GDH	4th generation of district heating
5GDH	5th generation of district heating
CHP	Combined heat and power
COP	Coefficient of performance
DH	District heating
DHC	District heating and cooling
EH	Excess heat
EU	European Union
GIS	Geographical information systems
IEH	Industrial excess heat
LCOHG	Levelized cost of heat generation
LAU	Local administrative units
MS	Member States (of European Union)
RES	Renewable energy sources
SCOP	Seasonal coefficient of performance
WtE	Waste-to-energy
WWTP	Wastewater treatment plants

conventional heat generation technologies such as CHP plants [4]. Lower operating temperatures inherently reduce heat loss during heat transportation and distribution. One of the most significant benefits of reduced system temperatures is unleashing the potential of low-temperature RES and EH. For instance, geothermal energy sources are available at different temperature levels depending on the geographical location. Similarly, EH is available at varying temperatures based on the process. Reduced DH supply temperatures enable exploiting low-temperature geothermal and EH sources, which cannot be utilized within the DH system with high supply temperatures.

In the quest to decarbonise the heating sector, policy direction plays a pivotal role, originating from the highest echelons of European governance. Through directives, the European Commission and Council chart the course for this transition. Key among these are the recently revised Energy Efficiency Directive EU/2023/1791 (EED III) [5] and the Renewable Energy Directive EU/2023/2413 (RED III) [6]. The EED III, with specific articles emphasising high-efficiency DH, and the RED III, set an indicative target of an annual 2.2 ppt increase in RES and waste heat and cold share from 2021 to 2030 in Article 24. Even though the target for DH is indicative, it gives a signal to the Member States (MS) that it might become binding in future amendments. Similarly, RED III Article 23 is amended to make the target of increasing the share of RES in heating and cooling by 1.1 ppt annually binding, which also puts the spotlight on DH collectively driving the agenda towards a more sustainable and efficient heating sector with an increasing focus on DH. These directives set a top-down framework that MS must adopt into their national laws. In this context, adopting a European-level perspective becomes crucial for comprehensively understanding and aligning with these overarching policy goals. Modelling at this level not only aids in assessing the collective impact of individual MS actions but also ensures that the strategic direction for decarbonisation is coherent, efficient, and aligned with the broader EU objectives. Such an approach enables a unified assessment of progress, challenges, and opportunities, guiding policy design and implementation across the EU for a harmonised transition towards sustainable heating.

Against this backdrop of policy-driven initiatives and the urgent need for a sustainable heating sector, our paper addresses the research questions.

- What is the impact of system temperature levels on the cost-minimal decarbonised DH supply mix in the EU-27 by 2050, considering the expansion of DH grids and the availability of RES and EH potentials?

- To what extent does the LCOHG decrease on the EU level by reducing DH system temperature levels?

These questions aim to contribute to understanding how DH system temperatures can influence DH systems' efficiency, sustainability, and alignment with EU climate objectives. By exploring this question, our research aims to provide a deeper insight into the dynamics of DH systems under varying temperature levels, assess the feasibility and implications of integrating diverse energy sources, and gauge the potential economic and environmental impacts.

Our work relates to two streams of studies in the literature: studies analysing the reduction of DH system temperatures and studies modelling the DH sector on a European level. In the former stream of studies, numerous studies focus on the different dimensions of the DH system temperature reduction as it is a complex topic requiring improvements on supply, building, and grid sides. Studies such as Refs. [7,8] show the monetary benefits of reducing DH system temperatures by calculating the cost reduction gradient for selected heat supply technologies and storage. Ref. [9] reviews DH business models with a particular focus on business models encouraging the reduction of return temperatures and provides suggestions for Austrian utilities. Ref. [10] investigates the effects of lower operational temperatures on a CHP-supplied DH system's overall energetic and exergetic efficiencies and costs. Ref. [11] evaluates the variable costs of DH plants under different supply temperatures. Ref. [12] analyses the network configurations implemented in low-temperature DH systems and discusses their suitability. Ref. [13] proposes a methodology to model the optimal retrofitting of DH distribution networks to lower the supply temperatures and implements it in a case study. Ref. [14] presents a method to estimate the potential temperature reduction in the existing DH substations. The implementation of the method in a case study shows that the supply temperatures on existing substations can be reduced significantly without any modifications. Ref. [15] compares low temperature and ultra-low temperature with electric boosting and heat pump boosting concepts for DH with a long-term energy system perspective. Through scenario analysis, the authors show that the low-temperature DH concept has the lowest costs. Ref. [16] provides a comprehensive review of benefits, challenges, required actions, and current practices of supply temperature reduction in existing DH networks. It is also essential to note that there are many studies in the literature focusing on 5GDH such as Refs. [17–19], however, 5GDH is out of the scope of this study.

A limited number of studies in the literature model the DH on a European level, mainly in the grey literature. Heat Roadmap Europe (HRE) project series provides local heat planning strategies to reach carbon neutrality by 2050 for selected MS, the highest 14 selected MS in the last series of the project [20]. The project series significantly contributes to understanding the decarbonisation of the heating sector from a European-level perspective. The sEnergies project presents pathways for EU-27 and each MS separately to reach carbon neutrality by 2050 with a focus on energy efficiency measures. Both studies employ the EnergyPLAN tool to simulate hourly energy systems with a whole energy system perspective [21]. Ref. [22] analyses the market value of renewable energy sources for EU-27, where DH is modelled as one of the flexibility options. Ref. [23] investigates the role of hydrogen and heat pumps in space and water heating for EU-27 in 2050, concluding that the heat pumps are more favourable over the hydrogen in both decentral and central heating systems. Both Refs. [22,23] employ the Enertile optimisation tool to model DH as a part of the European energy system. However, these studies do not consider the use of excess heat, and geothermal and solar generation are fixed, not optimised. Ref. [24] provides decarbonisation pathways for heating and cooling sectors of EU-27 by 2050. Our work builds on the modelling framework developed by Ref. [24]. However, our work still significantly differs from Ref. [24], as we design our primary scenarios based on temperature levels and provide a variety of sensitivity analyses.

This paper contributes to the existing literature in the following

ways.

- We model the DH sector for the EU-27 by designing the scenarios around the system temperatures, highlighting the explicit impact of temperature levels. This approach provides a unique insight into the operational efficiencies and potential of renewable energy integration under different system temperature levels.
- Our modelling considers a comprehensive range of DH supply technologies. To the best of our knowledge, our paper provides the future DH supply mix of EU-27 with the most detailed technological level by optimising their investments and operations.
- To the best of our knowledge, this study is the first to explicitly model the impact of supply temperature levels on the LCOHG of DH on the European level.

The structure of this paper is organised as follows: the next section describes the methodology employed in our study, including the data sources, key assumptions, and scenario design. Section 3 presents the findings, including the DH generation mix, installed capacities, LCOHG, and exploitation of RES potentials. Section 1 addresses the discussion and limitations of our study. Finally, Section 5 concludes the paper, synthesising the results and their policy implications.

2. Methodology

In this paper, we employ the Hotmaps DH Gen model [25] to model the future DH supply across EU MS. Our analytical framework focuses on two core system temperature scenarios: high-temperature and low-temperature DH, corresponding to 3GDH and 4GDH, respectively. We further investigate the sensitivity of specific variables like energy prices for biomass and hydrogen and the economic impact of implementing a carbon price on waste incineration.

2.1. Overview of modelling framework

DH, by its nature, shows heterogeneous characteristics within different regions. One of the factors creating this heterogeneity is the building stock. Depending on the current status and future evolution of the building stock, the feasibility of decentral heating or different DH generations and expansion of DH might change. Another critical factor leading to heterogeneity is the locally available heat sources. DH utilises locally available heat sources efficiently and effectively while helping secure the heat supply. Depending on the available heat sources, the dominant heat sources in DH supply vary in different regions. Both building stock characteristics and availability of local heat sources differ among and even within the MS. Thus, it is essential to consider these heterogeneous characteristics.

In our work, we reflect the heterogeneity of DH in the primary input data: DH demand and RES and EH potentials. This input data is prepared in three steps, where each factor creating heterogeneity is modelled in three steps via soft-linking three models. This framework is developed by the project Renewable Heating and Cooling Pathways – Towards full decarbonisation by 2050 (ENER C1 2019-482) [24] to model the DH ecosystem comprehensively. Fig. 1 provides an overview of the modelling framework, depicting data flow between models (in bold) and the DH system boundary. This paper primarily focuses on DH generation modelling, which is the DH Modelling System Boundary in Fig. 1. As it is essential to understand the broader context provided by the preceding phases of the modelling framework (steps 1–3), they are described briefly in the following paragraphs. The remainder of this section explains the preliminary work done to create data used in this paper that is already published in the respective papers and reports and allows us to capture the heterogeneity of DH systems. The last step of the modelling framework, DH modelling, is explained in the following subsection.

In the initial step, the INVERT/EE-Lab [26] model is employed by Ref. [24] to characterise the building stock. Invert/EE-Lab is a dynamic

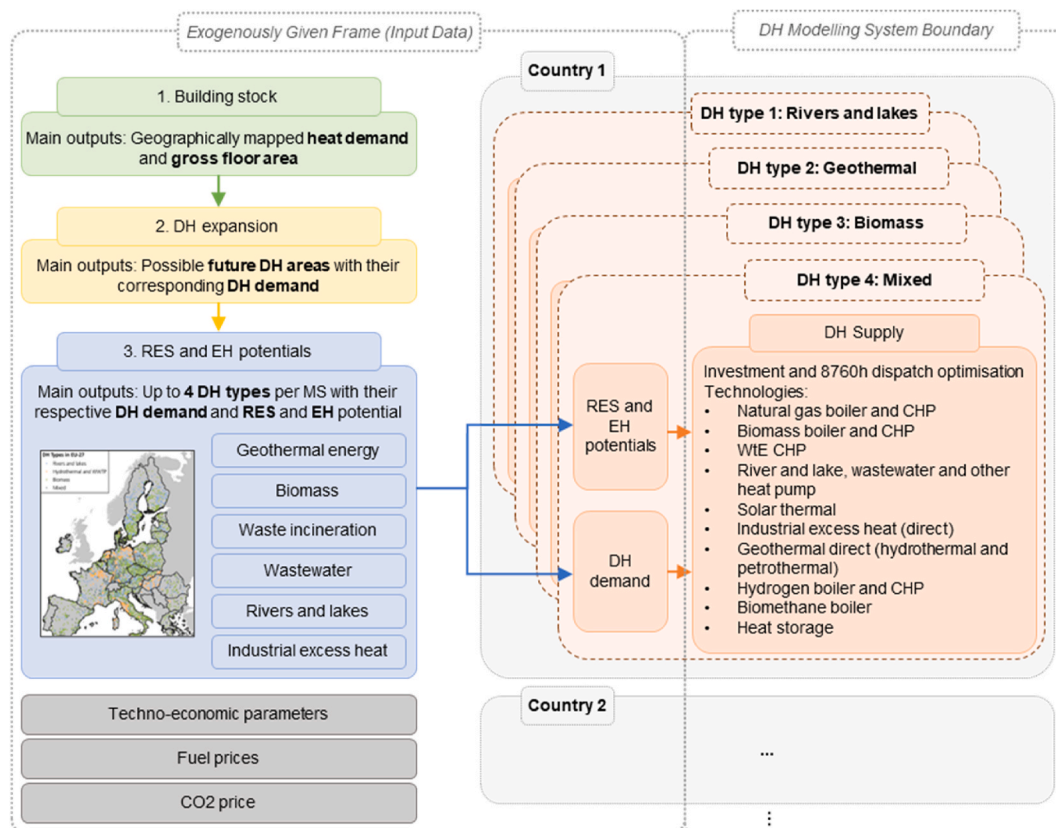


Fig. 1. Schematic overview of the modelling system boundaries and input data.

bottom-up simulation tool considering technological and socio-economic factors. It assesses how various policy combinations influence key aspects such as overall energy consumption, energy carrier mix, reductions in CO₂ emissions, and expenses associated with heating, cooling, hot water supply, and lighting in buildings. The primary outputs from INVERT/EE-Lab, which are utilized by the DH expansion model, are heat demand and gross floor area. These outputs are initially generated at the NUTS0 geographical level and further disaggregated to the hectare level for utilisation by the DH expansion model. A comprehensive description of the model setup and the results of the building stock modelling can be found in Ref. [24].

In the second step, the DH expansion model of Ref. [27] is employed by Ref. [24] to calculate potential future DH areas based on DH connection rates and grid costs through geospatial modelling. A DH region is designated as a potential DH area if its grid cost falls below a specified threshold (cost ceiling). These cost thresholds are assumed to be country-specific and the same within a country. Details about the grid cost ceilings and DH connection rates for each MS are available in Ref. [24]. The identified hectare-level DH areas are assumed to be connected if they lie in the same LAU regions. A total of 5815 prospective DH areas are identified. While temperature levels influence heat losses in DH areas, the regions and useful energy demand remain constant across the two temperature scenarios. Further information about model parameters and results can be found in Ref. [24].

In the third step, Ref. [28], building upon the first two steps, establishes spatially high-resolution datasets covering RES, including biomass, deep geothermal, heat from rivers and lakes, heat from WWTP, both from sewage water and biogas from sludge, and EH sources including IEH and WtE to quantify technical potentials for supplying the DH areas. The geothermal potentials are split up into hydrothermal (based on aquifers) and petrothermal (heat from the rock) potentials. We refer to the term geothermal as the sum of two types of geothermal energy unless one of them is explicitly stated. A detailed description of the methodology can be found in Ref. [28], and the open-source datasets are accessible through Ref. [29]. Assumptions regarding technical parameters and restrictions for each potential heat source to reflect the technical potential at two different temperature levels are established. Lower DH system temperatures enhance the efficiency and availability of heat sources, particularly geothermal energy. These supply potential datasets have spatial resolutions of point coordinates (WtE, WWTP) to 5000m × 5000m (rivers and lakes), varying depending on the heat

source, with an annual temporal resolution. The possible utilisation over the year is represented in terms of full load hours.

These supply potential datasets are spatially matched using a GIS-based approach, accounting for maximum distances for each heat source. The maximum distances are chosen for each source based on the heat that is typically available. The matching algorithm priorities areas with higher demand [28]. Thus, a single large point source could supply multiple DH areas, e.g., the WtE plant can supply multiple DH areas if one DH area's demand is lower than the heat potential. More details on maximum distances for heat sources and assumptions can be found in Ref. [28]. As a result, for each DH area, the technical potentials that are available to supply the DH areas from each heat source are listed as tabular data and spatial data in Ref. [29]. The DH demand and RES potentials for each MS are available in Appendix A.

A clustering analysis by Ref. [28] concludes the third step to identify so-called "DH types", defined as sets of DH areas with similar RES and EH potentials. Input data for clustering consists of the DH areas with their corresponding RES and EH potentials, normalised as percentages of demand to mitigate variations in magnitude (ranging from 0 % to 100 %) for each temperature scenario considered. DH areas are grouped into up to 4 clusters (DH types) for each MS using an agglomerative clustering algorithm [28]. Fig. 2 shows the geographical distribution of the DH types, named after the dominant heat source in each cluster, namely, Rivers and lakes, Hydrothermal and WWTP, Biomass, and Mixed.

2.2. DH generation model

Hotmaps DH Gen model is an open-source mixed-integer linear programming tool (available at Ref. [30]) that minimises the total cost of DH generation. The model optimises both investments and the operation of technologies on an hourly level. The model incorporates various input parameters including, but not limited to, hourly data related to district heating demand, electricity price, solar radiation, and source temperatures for heat pumps (river, wastewater, et cetera), as well as investment and operational costs. Each technology incorporated into the model is defined by its nominal heat production capacity and associated thermal and electrical efficiencies. The main outputs of the model are the levelized cost of heat production, associated investment, operational and fuel expenses, the distribution of heat production across different generators, CO₂ emissions, and full-load hours.

The objective function of the optimisation model and the primary

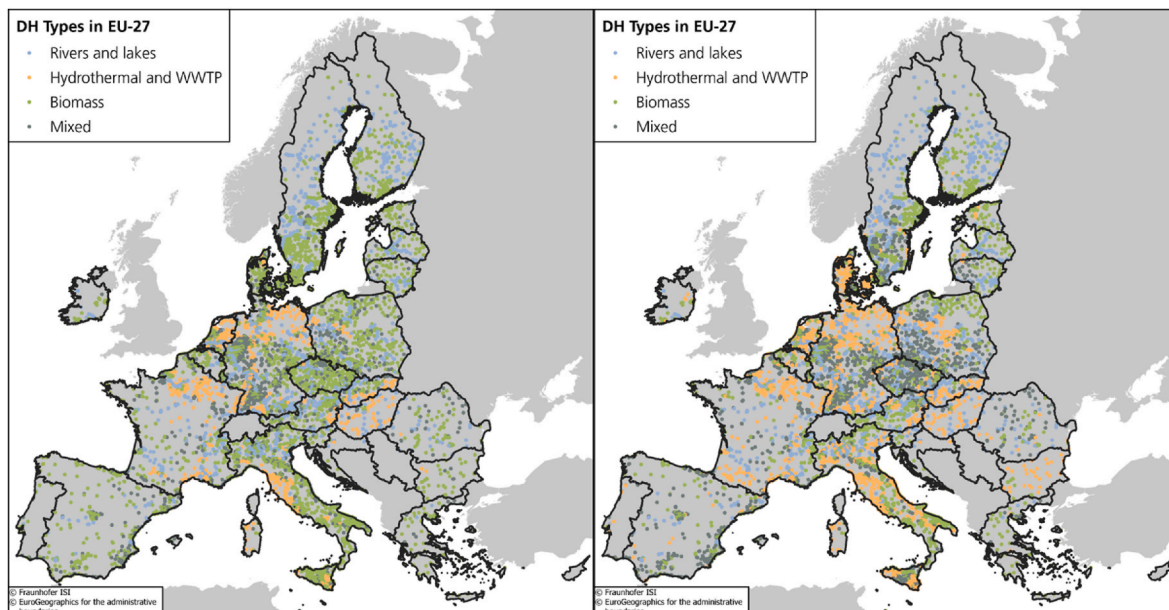


Fig. 2. DH areas in the EU identified by type in the high-temperature (left) and low-temperature (right) scenarios [22].

constraint to meet the DH demand are presented below. The model's complete mathematical formulation, which consists of numerous constraints, is available in Ref. [25].

$$\min(c_{total} - rev_{total}) \quad (1)$$

The objective function (1) of the model minimises the difference between the total cost of DH generation, c_{total} , and revenues from the electricity production, rev_{total} .

$$rev_{total} = \sum_{j,t} x_{elj,t} \cdot p_{s,elj,t} \quad (2)$$

The revenues from the electricity production (2) are defined as the sum of the electricity generated, $x_{elj,t}$, by each plant j at hour t times the electricity sale price, $p_{s,elj,t}$, for the plant j at hour t . Hence, it is assumed that all the electricity produced by the CHP plants is sold for the given electricity price.

$$c_{total} = OPEX_{fix} + OPEX_{var} + c_{cold} + c_{ramp} + IC \quad (3)$$

Equation (3) defines the total cost of heat production as the sum of fixed operational expenses, $OPEX_{fix}$, variable operational expenses, $OPEX_{var}$, cold start costs, c_{cold} , ramping costs, c_{ramp} , and investment costs, IC . The investment costs are annualised based on the interest rate and the lifetime of the heating plants (depreciation time).

$$\sum_{j,t} x_{thj,t} + \sum_{hs,t} (x_{unloadhs,t} - x_{loadhs,t}) = demand_{tht} \quad (4)$$

The demand constraint (4) ensures that the DH demand is met at all time steps, where $x_{thj,t}$ is the thermal generation by each plant j at hour t ; $x_{unloadhs,t}$ is the amount of thermal energy extracted from each heat storage hs while $x_{loadhs,t}$ is the thermal energy loaded in each heat storage hs at hour t . The sum of thermal generation by each plant and the thermal energy loaded and unloaded to/from heat storage must be equal to the thermal demand, $demand_{tht}$, at the corresponding time step t .

One of the most critical parameters of the optimisation model is the efficiency of the DH plants. Among the technologies considered in this study, only heat pumps have a variable COP value, while all the remaining technologies, including solar thermal, have a fixed nominal efficiency. Therefore, changes in the system temperature levels have a direct impact only on the heat pumps through their COPs. The COPs of the heat pumps are calculated as a function of the source, flow, and return temperatures.

$$n_{tht} = COP_{nomj} + SCOP_{source} \cdot (v_{sj,t} - v_{snomj}) + SCOP_{flow} \cdot (v_{fj,t} - v_{fnomj}) + SCOP_{return} \cdot (v_{rj,t} - v_{rnomj}) \quad (5)$$

Equation (5) defines the efficiency of heat pumps, n_{tht} , at each time step t . Nominal values for COP, COP_{nomj} , source temperature, v_{snomj} , flow temperature, v_{fnomj} , and return temperature, v_{rnomj} , are used as reference values. The nominal temperature values are subtracted from the actual source, flow, and return temperatures at each time step, represented as $v_{sj,t}$, $v_{fj,t}$, $v_{rj,t}$, respectively, and multiplied by the source, flow, and return temperature sensitivities, represented as $SCOP_{source}$, $SCOP_{flow}$, $SCOP_{return}$, respectively. In this way, the COPs of the heat pumps are pre-calculated and given the optimisation as a parameter. Similarly, the flow and return temperatures are also calculated as a function of the outside ambient temperature.

2.3. Data and assumptions

In our work, we model the target year 2050 by doing a so-called greenfield investment optimisation, meaning no plants exist at the beginning. The main reason for using a greenfield approach is that even though spatially high-resolution data is available for identifying prospective DH areas and local heat sources, data with similar granularity is lacking for current installed capacities and the remaining lifetimes of DH

plants, which is required to model the evolution of existing DH systems. We implement this approach with one exception: pre-installing capacities for WtE plants. This reflects the policy decisions as landfilling of waste municipal waste is forbidden in the EU, and incineration is the most favourable way of getting rid of the waste that cannot be recycled.

The projected DH demand for 2050 and the RES and EH potentials to meet this demand are foundational to our modelling study. Projections for 2050's heat and demand are derived from Ref. [28]. Ref. [28] implements stringent minimum energy performance standards for the worst-performing 25 % of buildings, mandating renovations within ten years starting in 2025. By 2035, these standards are tightened to improve energy performance to levels between classes A and B. Subsidies cover 15 % for standard renovations and 35 % for high-efficiency renovations, although pre-1945 buildings face barriers to achieving class A performance (see Ref. [27] for the details on policies implemented on building stock). The energy efficiency measures reduce the useful heat demand of the buildings from 3129 TWh in 2020 to 1900 TWh in 2050. DH final energy demand rises to 631 TWh compared to 446 TWh of DH final energy consumption in 2018 [3]. The DH shares of each MS as a result of the DH expansion modelling are available in Appendix B. We assume grid losses of 10 % in the low system temperature scenario and 16 % in the high system temperature scenario, which results in a DH demand of 695 TWh and 732 TWh, respectively. The difference in DH demand between the two system temperature scenarios comes from the higher grid losses due to the higher system temperatures. The RES and EH potentials are used as constraints on how much heat can be produced by the corresponding DH plants.

The techno-economic data used for DH plants is primarily based on Ref. [31], which provides a country- and size-specific dataset for the future cost of DH supply technologies. The authors use a construction cost index to scale the investment cost data within countries. However, they scale the investment cost as a whole even though the shares of equipment and installation costs in the investment cost are provided. This leads to big dispensaries within the countries. Thus, we adapt the investment cost data by keeping the equipment costs constant within countries and using the construction cost index only for installation costs, as proposed by Ref. [32]. The techno-economic data of Ref. [31] does not provide variable COP values for heat pumps but only SCOP values. Hence, we use temperature sensitivity values from Ref. [33] following the methodology presented in Subsection 2.2. Similarly, Ref. [31] does not cover the direct use of geothermal energy. Therefore, we use the techno-economic data for direct geothermal energy of Ref. [32].

Our analysis covers all relevant DH generation technologies, three of which are large-scale heat pumps. A complete list of technologies and how they are affected by system temperatures is available in Appendix C. We explicitly model wastewater and river and lake heat pumps. On the other hand, we model all other kinds of heat pumps, namely air-source, low-temperature geothermal and excess heat, and data centre heat pumps, aggregated into a single technology due to the lack of data on geographically dispersed low-temperature heat sources used by these heat pumps. We call this type of heat pump "other heat pumps" and model them as air-source heat pumps.

The energy carrier prices (including taxes and fees) and emission factors used in this paper are mainly based on Ref. [34]. The prices in Ref. [34], in particular electricity prices, are hourly prices for each MS. Ref. [34] uses a multisectoral energy supply model to model future electricity prices for a climate-neutral energy system, so that, among other things, interdependencies between sectors and the impact of high variability of RES are taken into account in the input data. Table 1 shows the overview of the energy carrier prices. Finally, we assume a high CO₂ price of 500 €/tCO₂ to ensure a decarbonised DH sector.

2.4. Scenario design

We frame our main scenarios based on the system temperature

Table 1
Overview of the energy carrier prices.

Energy Carrier	Price (average) [€/MWh]	Country-specific	Time Resolution	Source
Electricity	127	Yes	Hourly	[34]
Hydrogen	148	Yes	Annual	[34]
Biomethane	108	Yes	Annual	[34]
Biomass	35	No	Annual	Own assumption
IEH	10	No	Annual	Own assumption
Waste	0	No	Annual	Own assumption

levels: high- and low-temperature DH systems, representing the 3GDH and 4GDH, respectively. In our scenario design, the only difference between the 3GDH and 4GDH comes from the system temperature differences, even though the 4GDH covers more than only lower temperature levels, i.e., diversified heat sources, seasonal thermal storages, digitalisation, et cetera [35]. This makes modelling entire European DH systems computationally possible, and it also still aligns with the original definition of the 4GDH by Ref. [4]. The system temperatures affect the DH Gen model directly through the heat pumps and indirectly through the RES potentials. Table 2 gives an overview of the main scenarios and their temperature levels, which are in line with the temperature levels used in the calculation of RES and EH potentials.

In addition to the main scenarios, we calculate four sensitivities for each temperature scenario. We investigate the sensitivities of energy carrier prices for hydrogen and biomass and the investment cost of geothermal energy plants to reflect their uncertainties and assess the robustness of the results. The biomass price sensitivity reflects the uncertainty regarding its limited availability and competition with food production and other sectors as highlighted by Ref. [36]. This scarcity, coupled with its use in various renewable energy systems, makes biomass a sensitive and crucial factor in determining the economic feasibility of DH systems. Hydrogen prices were included due to the uncertainty surrounding its future role and evolving market conditions, which could significantly affect its competitiveness as an energy source. Geothermal investment costs were analysed due to the high initial capital required, influencing its adoption in low-temperature DH systems. Additionally, the increase in geothermal investment costs reflects the exploration risk, as deep geothermal projects typically involve multiple drillings, some of which may fail, resulting in higher overall investment costs. In our main scenario, we do not apply a CO₂ price to the WtE plants. Thus, in the last sensitivity scenario, we implement a CO₂ price for WtE plants to reflect the impact of corresponding policy decisions regarding monitoring WtE plants' emissions and evaluation of their inclusion into the emission trading system in the Revised Emission Trading Directive (EU 2023/959) [37]. Table 3 provides an overview of sensitivity scenarios.

3. Results

The presentation of our results is divided into three steps. Initially, we provide an overview of the DH generation mix for all scenarios with the corresponding LCOHG values, which cover only the heat generation costs (without grid costs). Subsequently, we present the installed

Table 2
System temperature levels used in the scenario design.

Scenario	Abbreviation	Flow Temperature	Return Temperature
High system temperature (baseline)	<i>HighTemp</i>	95–85 °C	60 °C
Low system temperature (baseline)	<i>LowTemp</i>	65–55 °C	30 °C

Table 3
Overview of the sensitivity scenarios.

Sensitivity Scenario	Abbreviation	Difference to the Baseline scenarios
High geothermal investment cost	<i>Geo</i>	100 % increase in geothermal investment costs
Low biomass price	<i>Bio</i>	Reduction of biomass price to €25/MWh
Low hydrogen price	<i>H2</i>	50 % decrease in hydrogen price
Carbon price for WtE	<i>WtE</i>	Implementation of a CO ₂ price for WtE plants

capacities for each scenario. Finally, we present the exploitation of RES and EH potentials in each scenario.

3.1. DH generation mix and LCOHG

This section analyses the DH generation mix by comparing the high and low system temperature scenarios and their corresponding sensitivities. Fig. 3 shows the DH supply mix of EU-27 in 2050 for all scenarios. The left y-axis shows the thermal generation in TWh, represented by the bars. The right y-axis shows the LCOHG in €/MWh, represented by the white and red squares. A more detailed figure on the DH supply mix of MS for the two main system temperature scenarios is available in Appendix D.

The HighTemp scenario primarily comprises heat pumps, accounting for nearly 50 % of the thermal generation (18 % from wastewater heat pumps and 31 % from other heat pumps). This is followed by 18 % from biomass (of which 16 % CHP), 12 % from WtE CHPs, and 13 % from direct geothermal energy sources (both hydrothermal and petrothermal). In the LowTemp scenario, the geothermal energy share significantly increases to 34 % and becomes the most dominant heat source together with heat pumps. Almost the entire DH production from wastewater heat pumps shifts to geothermal technologies, as well as half of the biomass share. One would expect an increase in the share of heat pumps in case of lower temperature levels; however, the results show that this is not necessarily the case due to the increased availability of alternative heat sources, such as low-temperature geothermal and excess heat, which become more feasible under reduced temperature levels.

Sensitivity scenarios were explored to better understand the cost effects, each offering its own set of key insights (cf. Table 3). In the LowTempGeo scenario, the share of direct geothermal energy use sharply declines to 16 %, and its share shifts to the heat pumps. There is no other significant change in the remaining technologies. This indicates a strong competition between heat pumps and the direct use of geothermal energy. However, in the HighTempGeo scenario, there is a minor change in the share of geothermal technologies. The limited availability of heat sources to supply high temperatures makes direct geothermal still an economically viable option even in case of high investment costs.

In low biomass price sensitivity, where biomass price drops from 35 €/MWh to 25 €/MWh, the contribution of biomass rises significantly in both temperature scenarios. In both scenarios, the increase in biomass shares originates from the drop in the heat pump shares. This implies a competition between heat pumps and biomass technologies.

In the case of the low hydrogen price scenario, the share of hydrogen increases significantly in both temperature scenarios compared to the baseline scenarios. However, the share of hydrogen in the overall DH supply mix remains low. It makes the highest contribution of 6 % in the HighTempH2 scenario.

In the last sensitivity scenario, the implementation of a CO₂ price for the WtE plants makes the share of waste incineration in the DH supply decline sharply in both temperature scenarios. Recall that the WtE capacities are pre-installed in our modelling (as explained in Subsection 2.3), which means that investment is already made into the WtE plants. This shows the strong impact of the high carbon price on the economic viability of waste incineration.

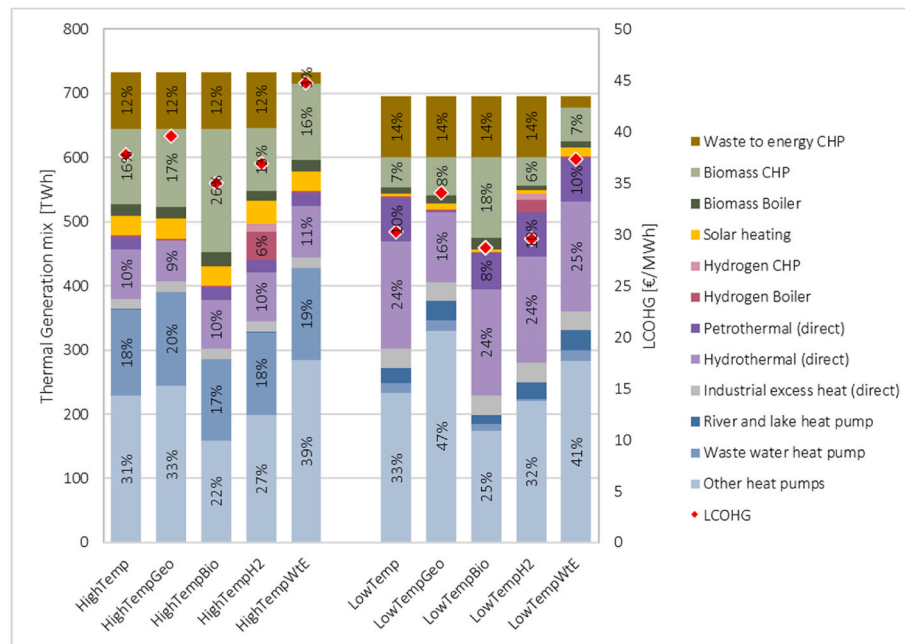


Fig. 3. DH thermal generation mix in the EU in 2050.

Wastewater heat pumps significantly contribute to DH supply across all high-temperature scenarios due to their relatively stable and high source temperature. However, their contribution plummets in all low-temperature scenarios as the availability of other heat sources, such as geothermal energy and IEH, increases, and river and lake and other heat pumps can also reach higher COPs.

Table 4 shows the LCOHG values for all high- and low-temperature scenario variants, the reduction in LCOHG in each scenario pair, and the total monetary savings. In terms of LCOHG, all low-temperature scenarios consistently demonstrate lower costs when compared to their high-temperature equivalents and a cost reduction of up to 20 %. This underscores the cost-effectiveness of transitioning to lower system temperature levels. In both temperature scenarios, the highest LCOHG occurs in case of a CO₂ price for waste incineration, and the lowest occurs in low biomass price scenarios. The results also show that reduction of system temperature can lead to annual savings of up to 6.83 billion Euros, which can be invested into measures required for temperature reduction.

3.2. Installed capacities in DH

Fig. 4 shows the DH installed capacities of EU-27 in 2050 across all scenarios. The installed capacities follow a similar pattern to the DH generation mix, with one notable exception: hydrogen boilers. While hydrogen boilers are among the highest installed capacities in all the scenarios, their contribution to the thermal generation mix is among the

lowest. This high capacity and low generation pattern underlines their role as backup technologies in peak demand hours.

On the other hand, geothermal energy technologies and WtE plants show an opposite trend. Despite their lower installed capacities, these technologies comprise a considerable portion of the DH supply mix. The reason for this disparity lies in their role as base-load suppliers, attributed to their stable heat supply and low operational costs. However, their deployment is constrained by higher investment costs.

3.3. Exploitation of RES and EH potentials

In this section, we analyse the exploitation rate of the RES and EH potentials, represented as the share of the utilized potential in the total potential for respective RES and EH sources. Fig. 5 visualises the exploitation rates in two heat maps for high-temperature and low-temperature scenario variants. While 100 % exploitation means that all the available potential of the respective RES or EH source is exploited, any share less than 100 % means there is still unused potential left.

The high-temperature scenarios reveal a substantial potential for hydrothermal and WtE, consistently reaching near-full or full exploitation across different economic conditions, underlying their robustness and critical role in high-temperature DH systems. This is mainly due to their ability to supply heat at high temperatures and the absence of other RES and EH potentials. Petrothermal energy, while significantly exploited in the *HighTemp* scenario, shows a marked decrease in scenarios with increased geothermal investment costs, indicating its sensitivity to economic factors.

Conversely, low-temperature scenarios exhibit a diverse exploitation pattern, which can be attributed as an advantage contributing to the security of supply through diverse local heat sources. Notably, there is a significant decrease in the use of wastewater treatment and petrothermal energies compared to high-temperature scenarios, suggesting lower competitiveness at reduced temperatures. In high-temperature scenarios, wastewater treatment plants are preferred due to their relatively high and stable source temperatures, resulting in a high COP, and a lack of other RES and EH sources. On the other hand, in low-temperature scenarios, the availability of hydrothermal energy increases significantly. Hydrothermal energy replaces wastewater treatment plants as a cheaper heat source due to its constant availability and

Table 4

LCOHG reduction and total annual savings across all the scenarios.

	LCOHG [€/MWh]		LCOHG Reduction	Total Annual Savings [bn EUR]
	HighTemp	LowTemp		
Baseline	37.75	30.24	20 %	6.64
Geo	39.62	34.07	14 %	5.35
Bio	35.05	28.69	18 %	5.74
H2	36.92	29.63	20 %	6.46
WtE	44.78	37.38	17 %	6.83

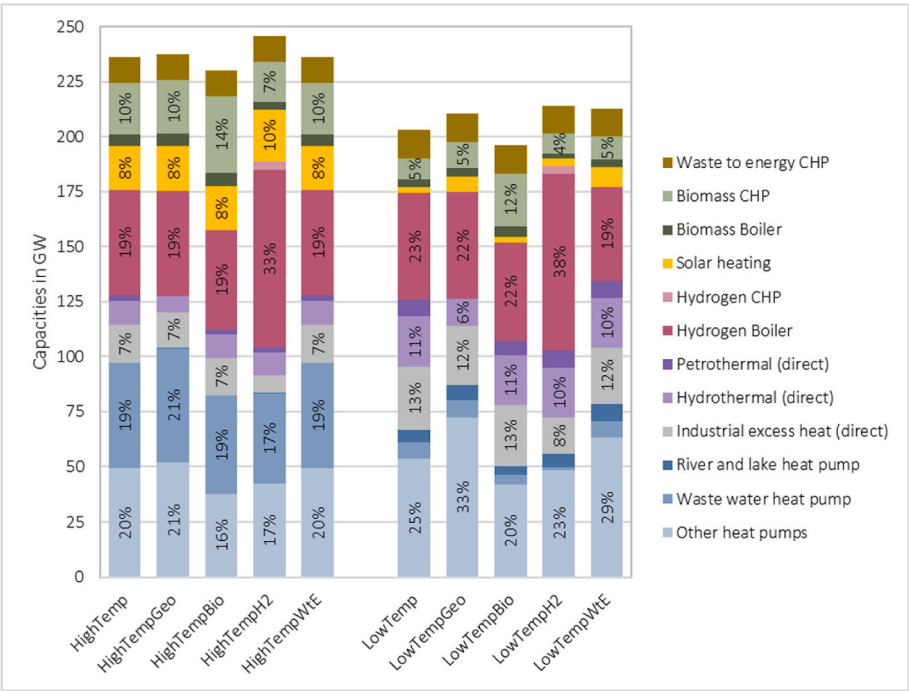


Fig. 4. DH installed capacities in the EU in 2050.

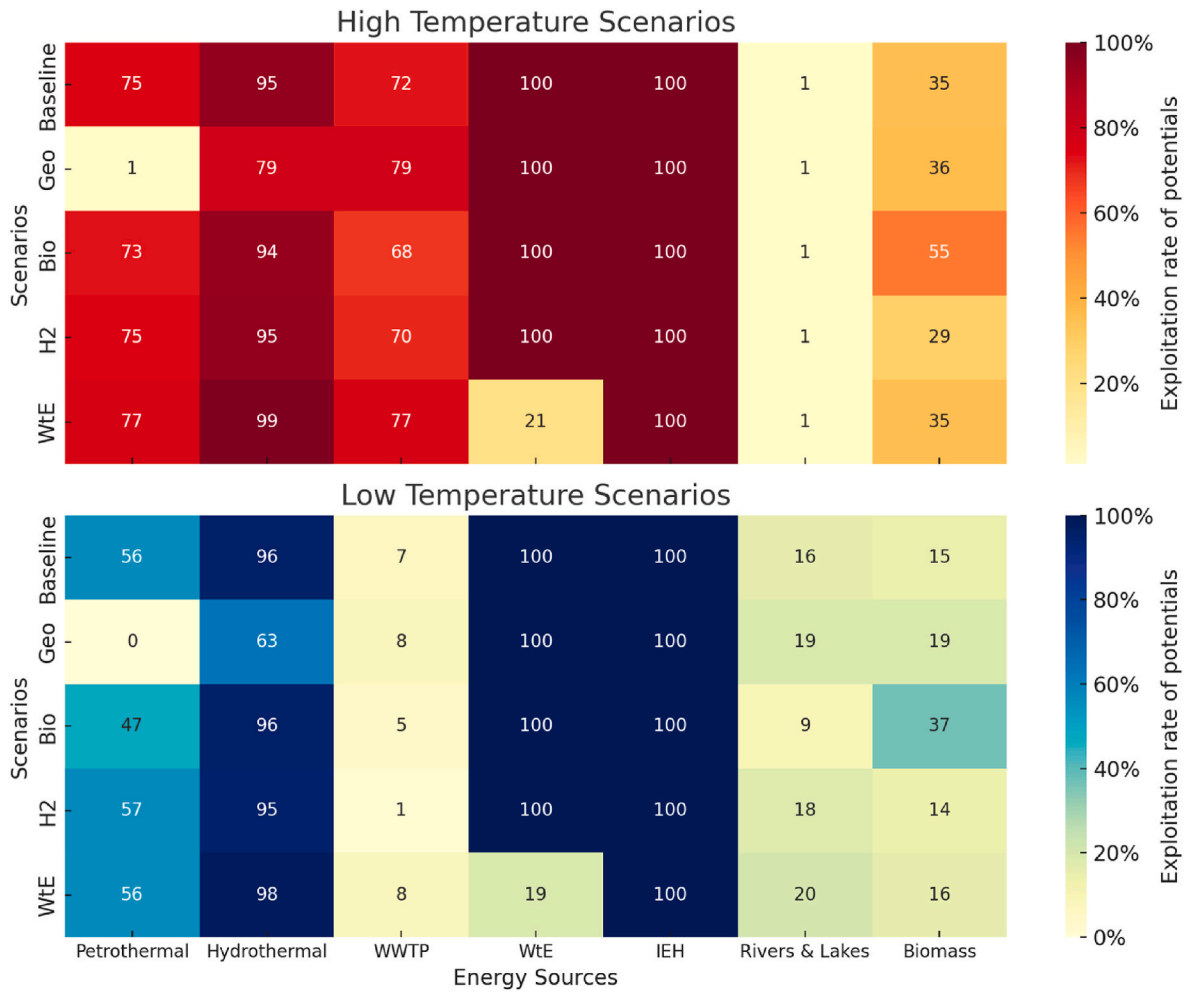


Fig. 5. Heat map of the exploitation of RES and EH potentials for high-temperature (top) and low-temperature (bottom) scenarios.

low operational costs.

IEH shows a consistent 100 % exploitation across all scenarios and temperatures, emphasising its significance and constant demand in DH systems. The notably higher exploitation of rivers and lakes in low-temperature scenarios shows that temperature reduction can unlock its potential.

The sensitivity scenarios introduce distinct shifts in the exploitation patterns. A notable increase in geothermal investment costs leads to a drastic reduction in petrothermal source utilisation, emphasising its economic sensitivity. The biomass price reduction and the hydrogen price decrease show an uptick in their respective resource exploitation, highlighting price as a significant driver for their adoption. The implementation of a CO₂ price for WtE plants resulted in a diversification of the supply mix, showcasing the potential for policy measures to influence sustainable energy exploitation.

4. Discussion and limitations

Our main results, the DH supply mix, are determined through the optimisation of capacity and operations. They rely on various techno-economic factors, including investment and operational costs, efficiencies, energy prices, operational flexibilities, and the RES and EH potentials. The available potentials ultimately determine the maximum amount of heat that can be generated by each heat source. Therefore, their competitiveness also depends on how much of the available potential is utilized. Subsection 3.3 and Fig. 5 give this perspective. Geographical constraints play a significant role in determining the potential of each heat source. This was calculated in the RES potentials by Ref. [28]. The authors match the available RES and EH potentials for DH with the DH areas defined by the DH expansion modelling. Geographical constraints have a notable impact on the availability of renewable heat sources. However, the DH regions do not differ within the two temperature scenarios in our study. What changes is the amount of available heat from a heat source as a result of temperature reduction. For example, geothermal energy becomes more cost-effective as a higher amount of geothermal energy becomes available at lower temperatures, which can be used as a baseload supply.

A key limitation lies in the greenfield optimisation assumption in our modelling. It is a simplification required mainly due to the lack of data (as explained in Subsection 2.3) not accounting for existing DH plants. This approach overlooks the influence of these existing infrastructures on the transition to newer systems. Even though there are 26 years until 2050 and most DH plants will complete their lifetimes, existing DH plants could still influence the future DH systems through lifetime extensions and the know-how and qualified workforce developed through them. On the other hand, the existing DH supply of EU-27 highly relies on fossil fuels, and these plants, anyway, need to be replaced by 2050. Furthermore, the political context within each country can significantly impact the pace and direction of DH development. Future research could address this limitation by incorporating more detailed data on the current DH infrastructure, including installed capacities, plant lifetimes, and retrofit potential, to provide a more nuanced transition pathway.

In our DH supply technologies, we do not have electric boilers. This is for several reasons: first, Ref. [34], where we get the electricity and hydrogen prices, models the generation and distribution of these energy carriers explicitly in an integrated energy system model; however, it does not consider electric boilers as an option. Hence, the interaction between electricity generation and the use of electric boilers is not captured. Since, we rely on the electricity and hydrogen prices from this study, introducing electric boilers would cause inconsistency. The use of electric boilers would require a system-wide perspective to determine whether it's more efficient to use cheap or excess renewable electricity directly with electric boilers, or to store it as heat in thermal storage (possibly much more efficiently via large-scale heat pumps), or to use it for hydrogen production (again another type of energy storage) for peak-time usage. Further research should investigate this question from

a system perspective on the European level.

Our findings indicate that, although hydrogen boilers are prominent in installed capacities, the contribution of hydrogen technologies to the DH supply mix is minimal, representing less than 0.5 % of total thermal generation (with around 50 full load hours). Given hydrogen's marginal role in our results, and since we use a pure DH model that only meets the heat demand, this study does not capture the potential competition between hydrogen boilers and CHPs. Future research should explore these technologies in use-case-specific scenarios to better understand their utilisation potential within DH systems.

Another limitation is the conservative assumption in modelling low-temperature geothermal, excess heat, and data centre heat pumps as heat pumps with the same COP as air source heat pumps. The COP of heat pumps utilising sources other than air is likely to be higher, potentially underestimating their efficiency. This conservative approach may not fully capture the potential benefits of these technologies. To overcome this limitation, future research should focus on generating high-resolution spatial data for low-temperature geothermal and excess heat sources. Such data could make the modelling of these low-temperature heat pumps possible.

In addition, the excess heat sources included in our input data by Ref. [28] involve IEH and excess heat from waste incineration plants. However, the excess heat that can be utilized by DH is not limited to these potentials. Future research should explore the excess heat potentials from emerging technologies such as Power-to-X (e.g., hydrogen production, methanation) and data centres, given that the market for these technologies is growing rapidly. These potentials could significantly enhance the DH supply mix, particularly considering the full utilisation of IEH across all the scenarios in our study.

In preparation of the RES potentials, Ref. [28] employs conservative assumptions regarding biomass availability and costs, focusing on high prices and only the local availability of secondary biomass resources. This approach might not accurately reflect the dynamic market conditions and the potential availability of more affordable biomass sources. If cheaper biomass becomes available for the DH sector, it could significantly alter the supply mix and the overall economics of DH systems. Future research should explore a wider range of biomass price scenarios and availability, including potential advances in biomass technology and supply chains, to better reflect future market conditions.

The imposition of a CO₂ price on WtE plants could make them economically infeasible and could reduce the emissions they are causing. However, this raises a further question on what happens to the no longer incinerated waste. Even though circular economy practices could reduce the amount of waste, this question requires further research. The economic pressure through CO₂ price could also incentivise WtE companies to invest in carbon capture and storage. Future research could explore alternative waste management strategies and the role of carbon capture technologies, offering a broader range of solutions for managing emissions from WtE plants.

The methodology employed in this study is highly replicable for other regions and contexts, as we use open datasets on renewable energy sources, and excess heat potentials with the open-source Hotmaps DH Gen model. Researchers can also adapt our approach to different geographic areas.

5. Conclusions

Our findings reveal that large-scale heat pumps and geothermal energy technologies could play a pivotal role in decarbonising the European DH sector. These technologies are particularly effective in low-temperature scenarios, as they can operate with greater efficiency and ease of integration with renewable sources. Lower system temperatures thus pave the way for integrating more geothermal energy and various types of heat pumps, ultimately reducing the LCOHG up to 20 % compared to the case with high system temperature levels.

Second, changes in system temperature levels significantly influence

the choice of energy sources in DH systems. For instance, wastewater heat pumps are more prominently featured in high-temperature scenarios, while the river and lake heat pumps and biomass are generally more favourable for low-temperature scenarios.

Third, geothermal technologies show considerable potential, specifically in low-temperature scenarios, where they can potentially replace or complement heat pumps depending on factors such as investment costs. This spotlights the significance of geothermal energy as EU MS strive to decarbonise their heating sector, especially considering the benefits of geothermal energy, such as low environmental impact (minimal land coverage, low noise levels), a stable and reliable energy supply, and low operational costs over the long term. However, it is critical to address the high initial capital costs associated with their deployment for geothermal technologies to become viable.

Fourth, our scenarios indicate that hydrogen technologies only play a marginal role in the future DH supply, even under reduced hydrogen prices. Therefore, their contribution to district heating remains limited as of now.

Fifth, the role of biomass in the DH supply is found to be highly sensitive to price, where it competes with large-scale heat pumps. The role of biomass also depends on policy interventions like biomass allocation within the different sectors. The low share of biomass in the baseline scenarios does not mean that it does not have an essential role in the decarbonisation of DH. Biomass could serve as a vital transitional fuel as it constitutes a considerable share of renewable DH supply today.

From a policy perspective, these findings point to several actionable steps. Targeted investments in geothermal and heat pump technologies should be prioritised, especially for low-temperature DH systems. Regulations might also need to encourage the transition from high to low system temperatures, such as building renovation, as lower temperatures enable the incorporation of more renewable and efficient technologies into the energy mix. Furthermore, considering the urgency of meeting the EU's 2050 carbon neutrality targets, implementing higher carbon prices on fossil fuel-based systems and WtE could accelerate the shift towards cleaner, more sustainable energy sources.

In summary, our study provides insights into how system temperatures, the integration of RES and EH, and the economic viability of various technologies interplay to shape the future of sustainable heating.

The insights gained offer a roadmap for policymakers and industry stakeholders, emphasising the need for technological innovations and robust policy frameworks to support the transition towards more sustainable and efficient DH systems.

CRedit authorship contribution statement

Ali K  k: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anna Billerbeck:** Writing – review & editing, Visualization, Validation, Resources, Data curation, Conceptualization. **Pia Manz:** Writing – review & editing, Validation, Resources, Conceptualization. **Lukas Kranzl:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT 4 in order to improve readability and language checking. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. RES and EH Potentials of Individual MS

The figure below shows the DH demand and RES potentials for each MS by Ref. [28].

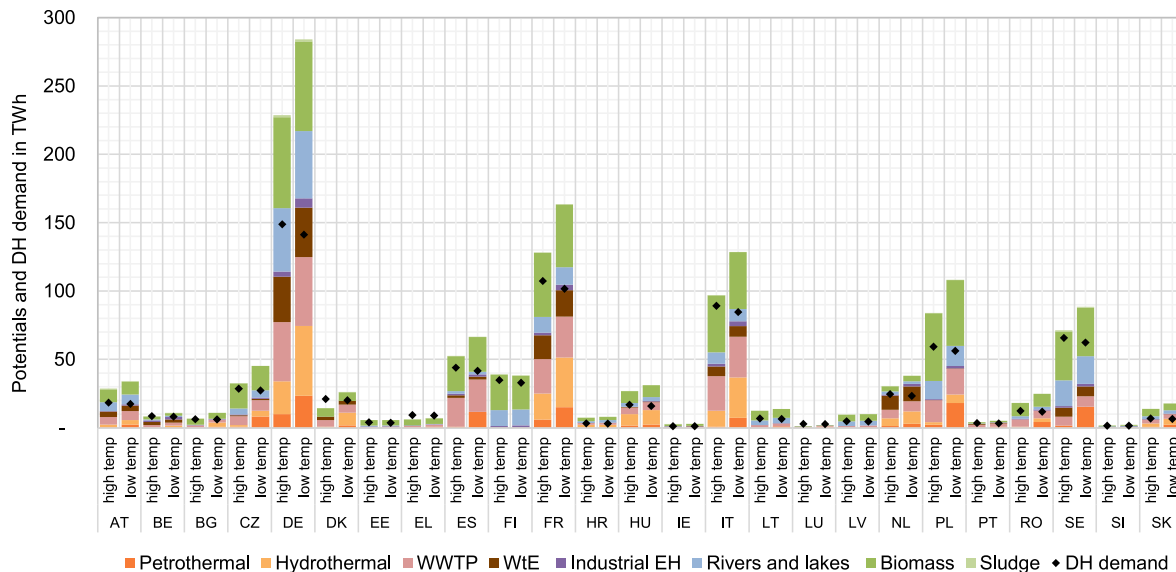


Fig. A-1. DH demand and RES potentials for each MS [28].

Appendix B. DH Shares

The table below shows the DH shares in each MS as a result of the DH expansion modelling by Ref. [24]. The details on the DH expansion model are available in Refs. [27,28].

Table B-1

Share of DH in the total heat demand [23,27].

Country	Demand in DH areas in GWh		Heat demand covered by DH in %		Heat demand covered by DH in GWh		Average distribution costs in €/MWh
	2020	2050	2020	2050	2020	2050	
AT	38,763	19,890	25.30 %	33.20 %	21,319	15,912	28.1
BE	26,027	10,636	3.40 %	14.10 %	3904	7445	32.8
BG	10,160	7382	26.70 %	34.90 %	6503	5536	28.7
CZ	51,729	30,994	24.70 %	43.00 %	22,761	24,795	28.1
DE	375,321	171,208	14.90 %	32.40 %	120,103	128,406	33
DK	35,273	20,287	52.80 %	46.70 %	31,040	18,259	32.6
EE	9105	4293	42.10 %	52.20 %	5281	3434	14.3
EL	14,346	11,582	10.80 %	24.80 %	4160	8107	29.4
ES	65,881	54,311	1.40 %	30.80 %	1976	38,018	33.3
FI	57,870	33,453	47.60 %	63.00 %	36,458	30,108	30.5
FR	203,160	123,460	6.30 %	27.50 %	30,474	92,595	33.7
HR	7702	3614	11.30 %	25.70 %	2850	2892	31.9
HU	32,249	18,281	13.30 %	32.50 %	10,642	14,625	28.2
IE	1760	1630	0.00 %	4.70 %	0	1141	33.1
IT	178,312	109,970	7.90 %	31.00 %	30,313	76,979	33.8
LT	10,987	6625	49.20 %	59.20 %	8570	5962	15.9
LU	5266	3066	19.50 %	51.70 %	1527	2453	29
LV	11,195	5253	41.50 %	59.90 %	6381	4203	15.9
NL	46,759	28,372	9.00 %	26.50 %	12,157	21,279	35.7
PL	130,002	63,942	27.20 %	38.00 %	66,301	51,153	29.6
PT	5437	4390	6.40 %	13.90 %	1794	3073	40.5
RO	29,236	14,314	15.00 %	26.00 %	12,571	10,735	29.8
SE	66,416	62,989	66.30 %	65.50 %	57,118	56,690	30.7
SI	3192	1809	13.60 %	19.20 %	1660	1447	30.7
SK	13,582	6614	32.70 %	36.90 %	10,051	5952	29.4
EU	1,429,732	818,367	16.20 %	33.20 %	505,916	631,202	31.4

Appendix C. DH Generation Technologies

The table below shows the list of DH generation technologies modelled by the Hotmaps DH Gen model in this paper and if and how they are affected by the reduction in system temperatures. While direct impacts mean the effect of temperature reduction is explicitly modelled in the DH generation modelling, indirect impact means the effect is reflected in the RES potential data.

Table C-1

List of DH generation technologies and the impact of temperature levels on them.

Technology	Impacted by Temperature Reduction	Type of Impact
Natural gas boiler	No	–
Natural gas CHP	No	–
Biomass boiler	No	–
Biomass CHP	Yes	Indirect through an increase in potential
WtE CHP	Yes	Indirect through an increase in potential
River and lake heat pump	Yes	Direct through an increase in COP of the heat pump
Wastewater heat pump	Yes	Direct through an increase in COP of the heat pump
Other heat pumps	Yes	Direct through an increase in COP of the heat pump
Solar thermal	No	–
Industrial excess heat (direct)	Yes	Indirect through an increase in potential
Geothermal direct (hydrothermal)	Yes	Indirect through an increase in potential
Geothermal direct (petrothermal)	Yes	Indirect through an increase in potential
Hydrogen boiler	No	–
Hydrogen CHP	No	–
Biomethane boiler	Yes	Indirect through an increase in potential
Heat storage	No	–

Appendix D. DH Generation Mix of MS

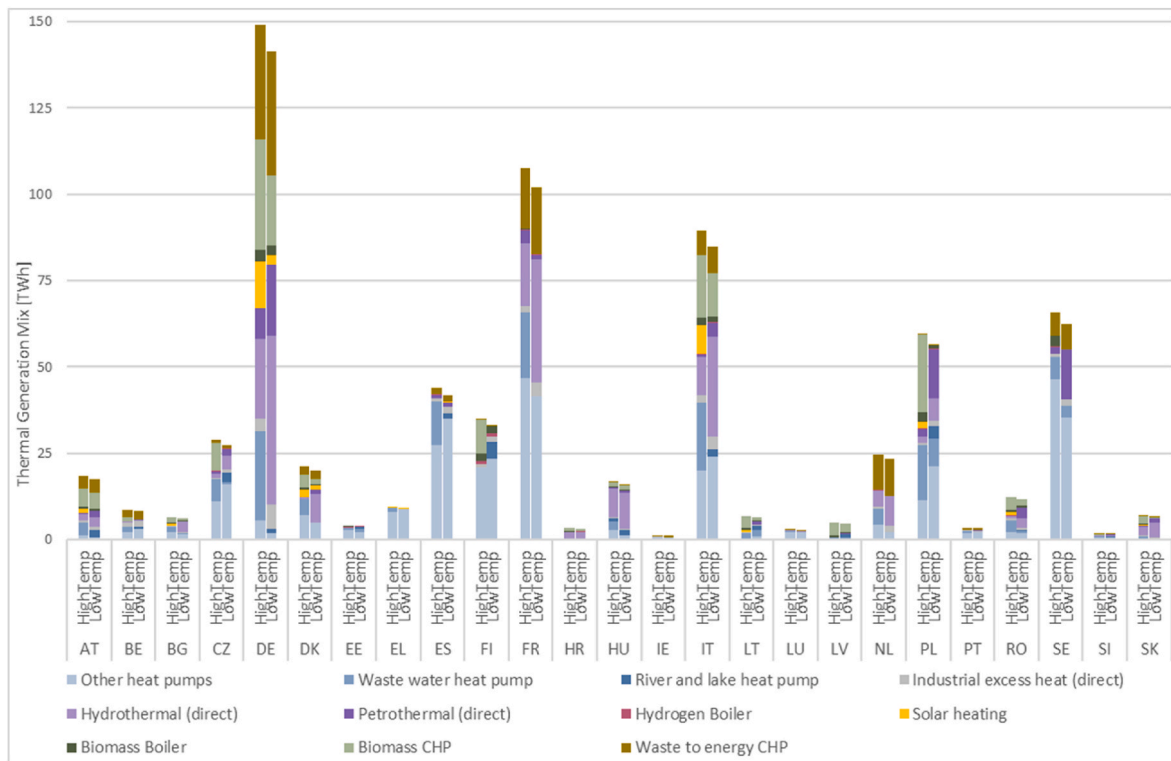


Fig. D-1. DH generation mix of MS for the main temperature scenarios.

Data availability

Data will be made available on request.

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